Pricing Derivatives Using Monte Carlo Simulation
Part 2

1 Reason of Bad Accuracy

When the strike price of an option increases significantly, we observe that the percentage error in option pricing becomes unacceptably high. This raises the question: what causes this issue in option pricing, and how can we improve the accuracy? To address this, we will employ a well-known technique called importance sampling.

Before introducing importance sampling, we first investigate why the accuracy deteriorates when the strike price of a call option becomes very high or the strike price of a put option becomes very low. To understand this, we consider the call option price using the replication technique. The value of a call option at time zero is given by:

$$V_0 = E\left[\max(S_T - K, 0)\right] = e^{-rT}E\left[\max(S_T - K, 0)\right], \tag{1}$$

where S_T is the stock price at time T, K is the strike price, r is the risk-free rate, and $E[\cdot]$ denotes the expectation under the risk-neutral measure. This can be expressed as:

$$V_0 = e^{-rT}\left[E\left(\max(S_T - K, 0) \mid S_T > K\right)P(S_T > K) + E\left(\max(S_T - K, 0) \mid S_T < K\right)P(S_T < K)\right]. \tag{2}$$

Since $\max(S_T - K, 0) = 0$ when $S_T < K$, the second term vanishes, simplifying the expression to:

$$V_0 = e^{-rT}P(S_T > K)E\left[\max(S_T - K, 0) \mid S_T > K\right]. \tag{3}$$

The probability $P(S_T > K)$ is straightforward to estimate using Monte Carlo simulation, as it is the expectation of the indicator function $E[1_{\{S_T>K\}}]$. However, the conditional expectation $E[\max(S_T - K, 0) \mid S_T > K]$ poses a challenge. This term requires generating stock prices S_T that are greater than K in the Monte Carlo simulation. Since S_T depends on Brownian motion, which follows a normal distribution, generating sufficiently large S_T values becomes increasingly difficult as K increases. This leads to poor accuracy in the Monte Carlo estimate, particularly for high strike prices. To address this issue, we propose using importance sampling to improve the efficiency and accuracy of the simulation.

2 Normal Distribution of Stock Price

To illustrate the challenge of generating stock prices $S_T > K$ in Monte Carlo simulations, we plot the normal distribution of S_T, with the mean at S_T and the strike price K positioned to the right.

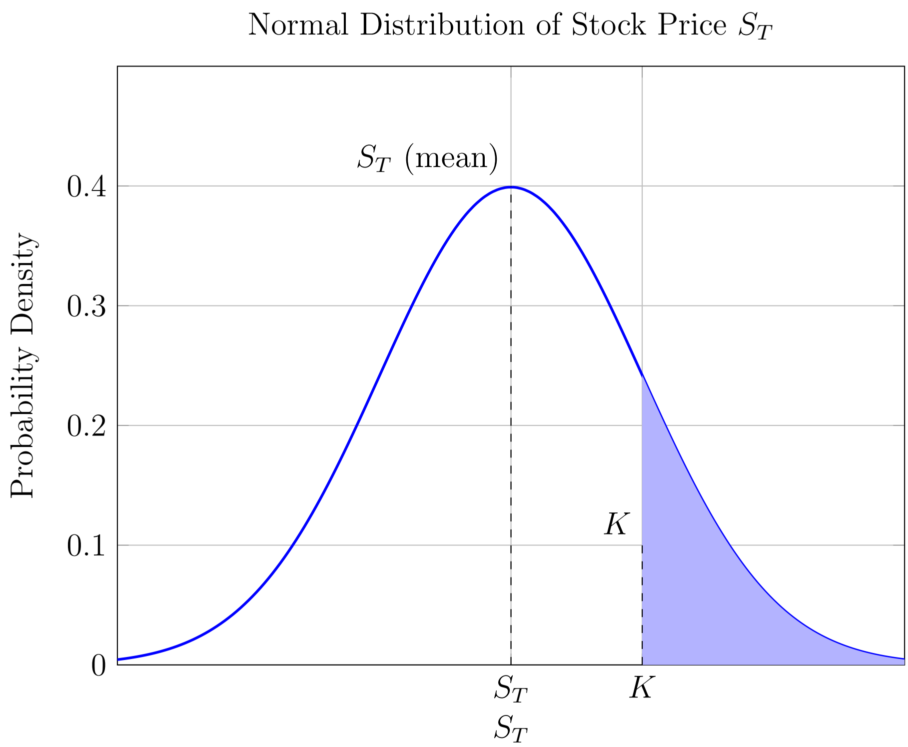

Figure 1: Normal distribution with mean at $S_T = 0$

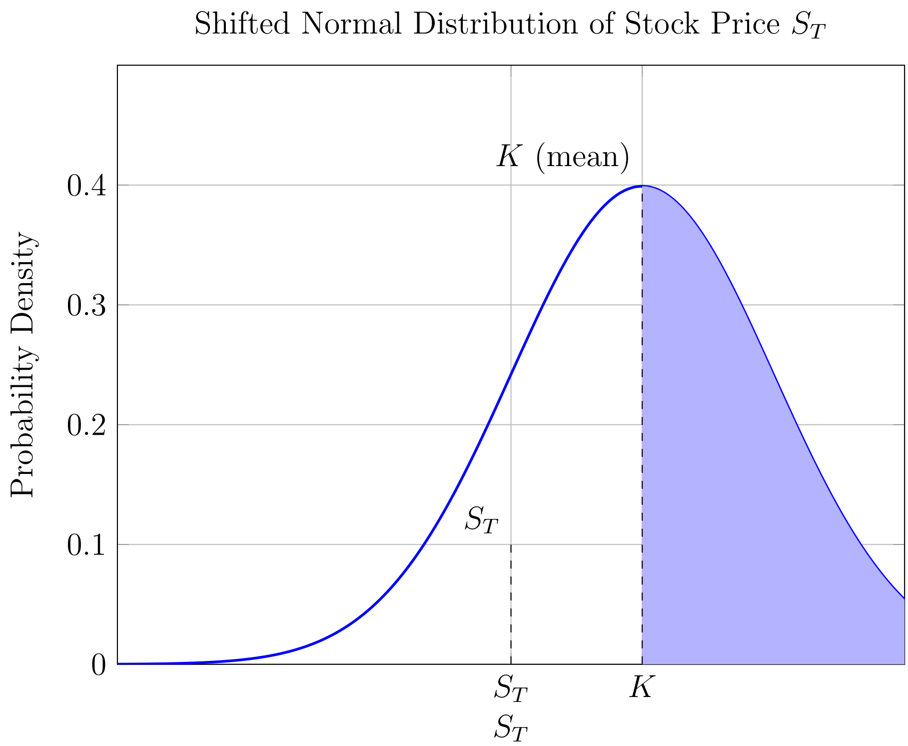

Figure 2: Shifted normal distribution with mean at $K = 1$

Figure 3: Top: Normal distribution of stock price S_T with mean at $S_T = 0$. Bottom: Shifted normal distribution with mean at $K = 1$. Both show the shaded area from K to infinity (approximated as $x = 3$), representing $P(S_T > K)$, which has a higher likelihood for large K when we try to shift the normal distribution to generate more data which S_T is greater than$K$.

3 Importance Sampling (Change of Measure)

The idea of importance sampling is that we do not have enough samples to ensure $S_T > K$ for a call option. Hence, we aim to generate a larger likelihood of obtaining $S_T > K$. Originally, $S_T \sim f_X(x)$, and the expectation $E_f[h(S_T)]$ is given by:

$$E_f[h(S_T)] = \int_{-\infty}^{\infty} h(S_T) f_X(x)\, dx. \tag{4}$$

To address this, we construct another probability density function $g_X(x)$ and rewrite the integral as:

$$E_f[h(S_T)] = \int_{-\infty}^{\infty} h(x) \frac{f_X(x)}{g_X(x)} g_X(x)\, dx. \tag{5}$$

Let $H_X(x) = h(x)\frac{f_X(x)}{g_X(x)}$, where $g_X(x)$ is our new probability density function. The expectation becomes:

$$E_f[h(S_T)] = \int_{-\infty}^{\infty} H_X(x) g_X(x)\, dx = E_g\left[h(x)\frac{f_X(x)}{g_X(x)}\right]. \tag{6}$$

By sampling from $g_X(x)$, which is designed to increase the likelihood of $S_T > K$, we can improve the efficiency and accuracy of the Monte Carlo simulation for option pricing. The classical pricing formula for S_T is $S_T = S_0 e^{(r-\sigma^2/2)T+\sigma W_t}$, where W_t is Brownian motion, which follows a normal distribution $\mathcal{N}(0, T)$. However, we now select $g_X(x)$ to also follow a normal distribution with $\mathcal{N}(\mu, T)$.

$$S_T = S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma W_t} \tag{7}$$

$$g_X(x) \sim \mathcal{N}(\mu, T) \tag{8}$$

Originally, when we attempted to use $n = 200,000$, the percentage error became excessively high. However, we now employ importance sampling, selecting an appropriate value for μ. The following presents the updated version of the option price and its standard error:

Table 1: Relation Between Strike Price K and the Error for $n = 200,000$ for call option after using importance sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.8818571310	0.0070251153	0.1194370280128
70	4.3016414843	0.0056079801	0.1303683739831
80	2.1986407229	0.0034892939	0.1587023231909
100	0.5246847674	0.5246847674	0.2380389366945
120	0.1178213526	0.0004209657	0.3572915221543
145	0.0179250022	0.0001055605	0.5889010725830

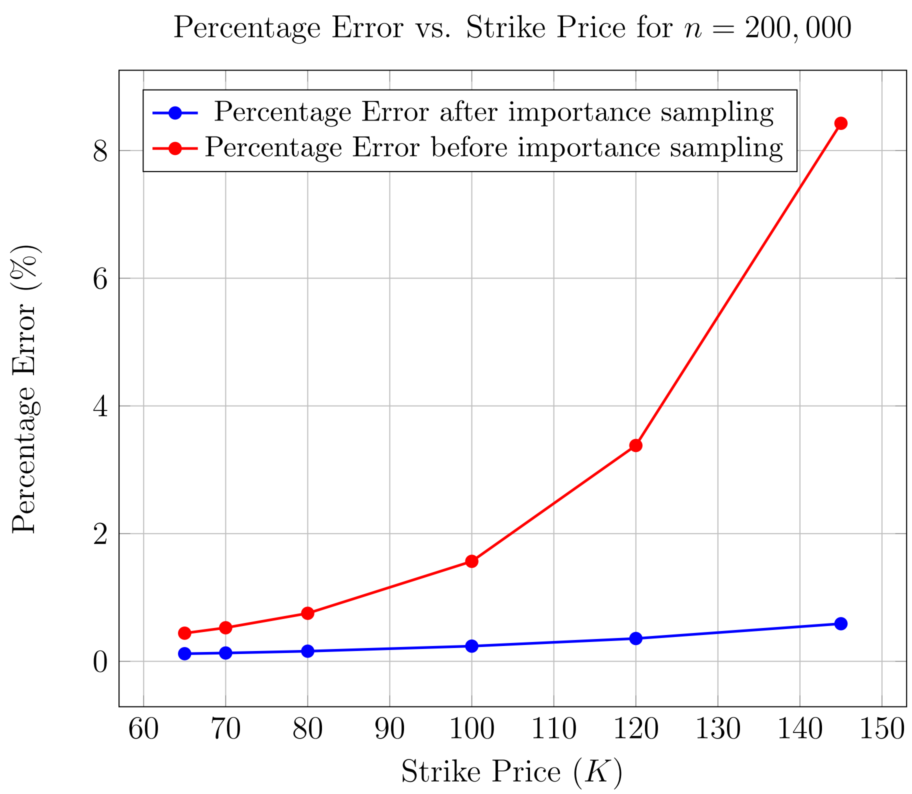

Figure 4: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$.

Table 2: Relation Between Strike Price K and the Error for $n = 200,000$ for put option after using importance sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9504380596	0.0043639532	0.1104675775744
50	2.2664069060	0.0028716932	0.1267068667896
40	0.4706504711	0.0008745451	0.1858162575359
30	0.0325958339	0.0001135192	0.3482628384181
25	0.0040788935	0.0000228700	0.5606901009596
20	0.0002017843	0.0000022716	1.1257547338531

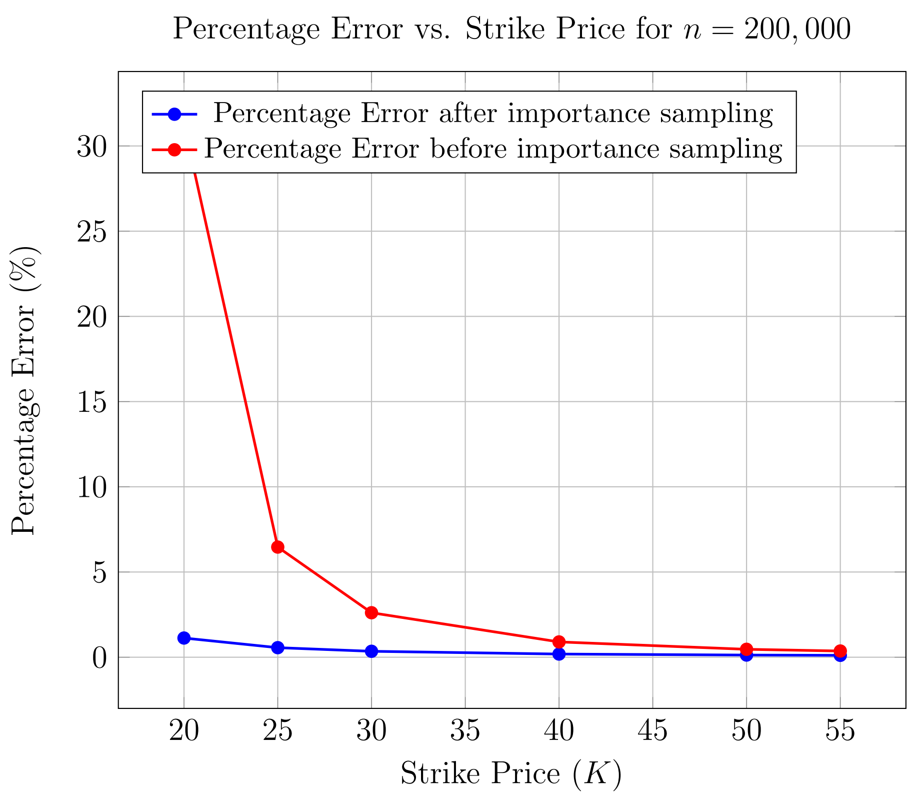

Figure 5: Relationship between strike price (K) and percentage error for Monte Carlo simulation with $n = 200,000$.

4 Path-dependent options using importance sampling

The above case compares the results to the original trial for $n = 1,000,000$. Now, we return to our case with $n = 1,000,000$ for the call option, put option, and Asian call option. We apply the same technique (importance sampling); for the call option, the table and graph below represent the option price and percentage error.

Table 3: Relation Between Strike Price K and the Error for $n = 1,000,000$ for call option after using importance sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	5.9014681454	0.0031470324	0.0533262623523
70	4.2964380803	0.0025093720	0.0584058699562
80	2.1992973271	0.0015607173	0.0709643624772
100	0.5246762827	0.0005583837	0.1064244289256
120	0.1181249456	0.0001883408	0.1594419875788
145	0.0181125664	0.0000474354	0.2618922154102

Table 4: Relation Between Strike Price K and the Error for $n = 1,000,000$ for put option after using importance sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
55	3.9512231202	0.0019514837	0.0493893584183
50	2.2672691020	0.0012846926	0.0566625555659
40	0.4700305078	0.0003907590	0.0831348149654
30	0.0328826044	0.0000510000	0.1550972568203
25	0.0040565666	0.0000102098	0.2516859715651
20	0.0002018025	0.0000010149	0.5028973426857

Table 5: Relation Between Strike Price K and the Error for $n = 1,000,000$ for Asian call option after using importance sampling

Strike Price (K)	Option Price	Standard Error	Percentage Error (%)
65	2.5645609853251665	0.0037335023013196416	0.14558056223592836
70	1.370926588692837	0.002004978094734432	0.14624985110589734
80	0.333514790605139	0.0006583680852701578	0.1974029649706376
100	0.013667260376728172	0.00006185267663825915	0.4525609005267679
120	0.00045757335722987623	0.000005326766043550572	1.164133785192066
145	0.000006775516222498219	0.0000004524948221504276	6.678381503211678